Network Intrusion Detection System (IDS): An Explainable Machine Learning Case Study

Framework: Python | Scikit-learn | Streamlit | Pandas | Plotly
GitHub: https://github.com/CyrixX0001/network-intrusion-detection

1. Problem Statement

Modern enterprise networks generate massive volumes of packet-level telemetry every second. Traditional signature-based intrusion detection systems (such as legacy Snort or Suricata rulesets) rely heavily on known exploit signatures and fixed rule patterns. While effective against known threats, signature-based approaches exhibit critical vulnerabilities:

- Zero-Day and Polymorphic Attacks: Novel exploits evade static signatures.
- High Operational Overhead: Maintaining signature databases requires continuous human intervention.
- Volume and Velocity: Volumetric attacks require adaptive statistical anomaly detection.

A Machine Learning-based Intrusion Detection System (ML-NIDS) addresses these challenges by learning multivariate behavioral distributions from connection metadata (packet size, duration, flags, login retry counts). The goal is to accurately classify Normal Traffic (0) vs Malicious Intrusions (1) with low false alarm rates.

2. Algorithm Justification: Why Random Forest?

- Overfitting Resistance: Averages multiple de-correlated trees via bagging to eliminate variance.

- Non-

3. Telemetry Features Overview (10 Attributes)

1. packet_size (Bytes) | 2. connection_duration (s) | 3. failed_logins (Count)
4. bytes_transferred (KB) | 5. src_to_dst_ratio | 6. urgent_packets (Flags)
7. wrong_fragments (Anomalies) | 8. num_compromised | 9. root_shell_attempts | 10. hot_indicators

4. Setup, Installation & How to Use the Demo

1. Installation: pip install -r requirements.txt
2. Automated Tests: pytest test_model.py -v (all 4 unit tests passing)
3. Launch Web App: streamlit run app.py (accessible on http://localhost:8501)
4. Sidebar Controls: Configure number of trees (10-300) and max depth (2-25), click 'Retrain Model'.
5. Metrics & Plots: Review Accuracy, Precision, Recall, F1, interactive Confusion Matrix, and ROC-AUC curve.
6. Live Packet Testing: Use sliders for the Top 4 features or presets (Benign, DoS/Flood, Brute-Force) to inspect real-time threat predictions.